ReneWind V2 - Intelligent Predictive Maintenance Platform

Physics-Inspired Operational Constraint Network & Risk Intelligence

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Team Contributions & Presentation Guide

For the presentation, our team of 4 has divided the workload and report sections as follows. Each member will present their respective domain:

- **Member 1 (Data Engineering & EDA):** Led data ingestion, cleaning, scaling, handling class imbalance, and exploratory data analysis. *(Primary Speaker for Sections 3, 4, 5)*
- **Member 2 (Deep Learning Architect):** Designed and trained the Physics-Inspired Operational Constraint Network (PIOCN) and implemented the custom Operational Consistency (OC) loss function. *(Primary Speaker for Sections 6, 7)*
- **Member 3 (Risk & Decision Intelligence):** Engineered the per-turbine adaptive thresholding mechanism, the Combined Risk Index, and the Greedy Fleet Dispatch Scheduler algorithm. *(Primary Speaker for Sections 10, 11, 12)*
- **Member 4 (Full-Stack & MLOps Engineer):** Built the FastAPI REST backend, integrated the SHAP explainability engine, and developed the Next.js 14 interactive operational dashboard. *(Primary Speaker for Sections 8, 9)*

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1. Introduction & Business Context

Wind turbine generator failures cause catastrophic breakdowns and unexpected grid downtime. In industrial wind farm operations, maintenance costs are severely asymmetric. A missed failure (False Negative) can result in an emergency replacement costing $200,000. Conversely, a false alarm (False Positive) triggers an unnecessary inspection costing $5,000.

Because the FN:FP cost ratio is 40:1, standard classification metrics like accuracy are inadequate. ReneWind V2 is designed to optimize directly for Recall and Total Dollar Cost, generating millions of dollars in annual net savings by identifying turbines at risk of failure before breakdown occurs.

2. Problem Statement

The goal is to develop a robust predictive maintenance pipeline that minimizes total operational costs by maximizing the early detection of wind turbine failures (Recall) while maintaining an acceptable false positive rate. The system must also incorporate operational constraints, quantify uncertainty, and provide actionable intelligence (e.g., dispatch scheduling) to operators.

3. Data Description *(Led by Member 1)*

The dataset consists of anonymized SCADA (Supervisory Control and Data Acquisition) telemetry data from wind turbine generators.

- **Features**: 40 continuous sensor variables (labeled `v1` to `v40`). These represent four broad physical categories (Vibration, Thermal, Pitch, Grid).
- **Target**: Binary classification (`target`). `Class 0` represents healthy operation, and `Class 1` represents a turbine failure.
- **Volume**: A training dataset (`Train.csv`) with 20,000 rows and a holdout testing dataset (`Test.csv`) with 5,000 rows.
- **Imbalance**: The dataset is highly imbalanced, with a failure rate of approximately 5.6%.

4. Exploratory Data Analysis (EDA) *(Led by Member 1)*

4.1 Data Overview

The dataset contains 20,000 records in the training set and 5,000 in the test set. All 40 predictor variables are continuous numeric (float) types. The target class is severely imbalanced (94.4% healthy vs. 5.6% failure).

4.2 Univariate & Bivariate Analysis

- **Vibration Sensors (e.g., v1, v4)**: Show right-skewed distributions with long tails. Outliers in these features are heavily concentrated in the positive (failure) class.
- **Bivariate**: Plotting individual features against the target reveals that no single feature acts as a perfect discriminator. Features like `v1` and `v24` show statistically significant median shifts, but the interquartile ranges overlap significantly. This indicates that univariate thresholding will yield high false positive rates.

4.3 Multivariate Analysis

- **Correlation Heatmap**: The correlation matrix reveals weak to moderate linear correlations between the 40 features. Most features have absolute Pearson correlation coefficients with the target below 0.15.
- **Interpretation**: The lack of strong linear predictors implies that the failure signature is highly non-linear and relies on complex interactions between multiple sensor subsystems.

4.4 Key EDA Insights (Summary)

- **Extreme Class Imbalance (5.6%)**: The model will naturally bias toward predicting the majority class unless explicitly penalized or rebalanced.
- **Weak Individual Predictors**: No single feature strongly correlates with the target, necessitating complex, non-linear ensemble or neural architectures capable of capturing interaction effects.

5. Data Preparation *(Led by Member 1)*

- **Missing Value Handling**: Utilized **Median Imputation** because several sensor distributions exhibit skewness and outliers.
- **Feature Scaling**: All 40 features were scaled using **StandardScaler** (Z-score normalization). This is critical for the neural network architecture to ensure stable gradient updates.
- **Class Imbalance Handling**: Applied **Class Weights** during training. Given the 40:1 cost asymmetry, the loss function heavily penalizes false negatives, forcing the network to prioritize recall.
- **Train/Validation Split**: A **Stratified Split** was used. Data leakage was prevented by strictly fitting the imputer and scaler *only* on the training split.

6. Model Building and Experimentation *(Led by Member 2)*

6.1 Baseline Model

- **Model Type**: Standard Multi-Layer Perceptron (MLP) / Base Neural Network.
- **Results**: Achieved high accuracy but poor recall. The network collapsed into predicting the majority class to minimize standard Binary Cross Entropy (BCE) loss.
- **Limitations**: Ignored physical constraints, leading to false negatives on borderline cases.

6.2 Additional Models (PIOCN)

- **Architecture**: Physics-Inspired Operational Constraint Network (PIOCN)
- **What Changed**: We introduced a custom loss function: `Total Loss = BCE + lambda * mean(OC_violation * (1 - p))`. We also integrated Monte Carlo (MC) Dropout during inference.
- **Hypothesis**: By pre-calculating a 3-sigma Operational Envelope strictly from healthy data, we can penalize the network if it predicts "Healthy" while the input sensors physically violate the normal operating envelope.
- **Results**: Recall jumped significantly while maintaining a manageable FP rate.

6.3 Model Evaluation Criteria *(Led by Member 3)*

- **Optimization Metric**: **Total Dollar Cost** and **Recall**.
- **Rationale**: Accuracy is deeply misleading here. A naive model predicting "Healthy" 100% of the time achieves ~94.4% accuracy but misses all failures, incurring maximum financial penalty.

7. Model Comparison and Final Model Selection *(Led by Member 2)*

- **Final Model Selected**: PIOCN.

- **Performance**: Achieved **91.9% Recall**, successfully catching the vast majority of impending failures.
- **Robustness**: MC Dropout provides variance (uncertainty) estimates for every prediction, making the model highly robust to out-of-distribution sensor noise.

8. Automated Pipeline / MLOps *(Led by Member 4)*

- **Pipeline Stages**: The project features a modular execution pipeline.
- **Tooling**: We utilize `uv` for ultra-fast deterministic dependency management, ensuring exact environment replication. The `ModelService` operates as a pre-warmed thread-safe singleton, caching model weights and scalers in memory to guarantee sub-200ms API response latencies.

9. Model Deployment and Inference *(Led by Member 4)*

- **Interface**: A production-grade **Next.js 14 App Router** web dashboard interfacing with a **FastAPI** REST backend.
- **Functionality**: The platform exposes 9 REST endpoints. The dashboard features a "Live Prediction Simulator", allowing users to adjust sensor sliders and instantly visualize the predicted failure probability, Risk Tier, and SHAP explanations.

10. Business Impact and Recommendation *(Led by Member 3)*

"For every 1,000 turbines monitored, assuming a 5.6% failure rate (56 failures), our model catches 51 true positives early (saving $10.2M), while generating 50 false alarms (costing $250K). Missing 4 failures costs $800K." **Net Annual Savings**: **$8,931,250** per 1,000 turbines compared to a purely reactive maintenance strategy. **Recommendation**: Deploy the model in "shadow mode" alongside current SCADA monitoring systems. Prioritize crew dispatches according to the built-in Fleet Scheduler.

11. Future Enhancements *(Led by Member 3)*

- **Temporal Modeling**: Transition from cross-sectional MLP architectures to LSTM or Transformer-based sequence models to capture temporal degradation patterns over sliding windows.
- **Drift Monitoring**: Implement automated data drift detection to trigger retraining when seasonal weather patterns alter sensor distributions.

12. Conclusion *(Led by Member 3)*

ReneWind V2 successfully demonstrates that injecting domain-specific physical constraints into deep learning loss functions yields vastly superior industrial outcomes compared to naive classification. By optimizing directly for the asymmetric financial costs of wind turbine maintenance, we engineered a system that maximizes recall and saves millions. The most important takeaway is that static thresholds and standard BCE loss fail in high-stakes environments; adaptive thresholds driven by uncertainty and operational constraints are mandatory for safe, cost-effective predictive maintenance.